

Social Affordances for “Playgrounds for Dialogue”: An Annotated Portfolio of Museum Installations Aimed at Social Connections

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Abstract

Museums are increasingly becoming participatory spaces. Nina Simon describes participatory museums as “meeting grounds for dialogue,” foregrounding their role in fostering connections among visitors. Motivated by this view, our design research explores how playful technology can support that move. In this poster paper, we introduce our ongoing work toward characterizing (and making actionable) a novel genre of museum-related technology design that we call “playgrounds for dialogue.” We present our in-progress analysis of an annotated portfolio of museum installations that support collective activity through play and playfulness, from which we identify three social affordances that can support playful forms of social connections among visitors around museum content, and begin to unearth key design considerations. Our in-progress contribution begins to flesh out “playgrounds for dialogue” as a *strong concept* and shows promise for guiding designers designing in this space. Key to these efforts is opening our meaning-making to the broader HCI community.

CCS Concepts

• Human-centered computing; • Interaction design; • Interaction design theory, concepts and paradigms;

Keywords

Social affordances, Play, Strong concept, Museum participation, Annotated portfolio

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1 Introduction and Background

Museums are multicultural public spaces that should foster respect for cultural differences and promote intercultural dialogue [4] [5] [21]. As organizational structures and functions evolve, museums are becoming participatory spaces “where visitors can create, share,

and connect with each other around content” [26]. Simon [26] describes participatory museums as “meeting grounds for dialogue,” encouraging institutions to collect and share diverse, personalized, and evolving content co-generated with visitors, and to invite visitors to respond to and discuss that content. This shift motivates us, as Human-Computer Interaction (HCI) researchers, to think about how visitors might engage socially around museum content, and how interactive technology might support meaningful collective participation and bonding within the premises of a museum. We aim to bring “meeting grounds for dialogue” [26] into HCI, focusing our investigation on how play and playfulness can support social engagement in audience experiences to promote the development of participatory museums.

The potential of play as a catalyst to facilitate meaningful social connections has been widely explored in HCI (e.g., [1], [11], [20]). However, in the context of HCI museums, most research has focused on more tangible, utilitarian outcomes of play or game, such as promoting learning (e.g., [7]). Our work highlights play’s *eudaimonic* capacity [24], i.e., its potential to be socially, emotionally, and culturally productive, even if it does not lead to tangible outcomes. These, too, are ambitions that participatory museums embrace, insofar as they are seen as spaces for community building, for bonding, for individual and collective growth [26]. In that regard, we argue that play’s capacity to promote social connections among museum audiences is relevant; for example, playfulness could go beyond pure entertainment to provide delightful museum experiences and opportunities for communication among others [14].

Building on this, we have a design research agenda to explore how playful technologies can support Simon’s vision of museums as “meeting grounds for dialogue” [26]. We refer to this emerging design space as “playgrounds for dialogue,” i.e., designing playful technology-mediated installations aimed at fostering social connections among people and collective engagement in museum-related spaces. In this paper, we present our ongoing efforts to flesh out “playgrounds for dialogue” as a *strong concept* [10] and try to make it actionable for designers. To do so, we offer in-progress findings from analyzing existing museum installations, in the form of an *annotated portfolio* [9][13] of design exemplars that reflect what “playgrounds for dialogue” might be. Looking closely at what these exemplars do and how they draw people into spontaneous, joyful collective behavior, we illustrate what this novel design space might look like, begin to elucidate its bounds, and identify diverse ways it might manifest.

Our in-progress contribution foregrounds a set of social affordances that might sustain “playgrounds for dialogue” and begins to



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Figure 1: Our iterative cycles of analysis through reflexive conversations about the design exemplars.

unearth important considerations for designers when envisioning such installations. Overall, with this paper, we aim to share the current state of our efforts toward fleshing out the strong concept of “playgrounds for dialogue” and to show the promise of providing guidance for designers interested in this genre of playful technology. In future stages of our work, we will consolidate our analysis and explore how to make the resulting insights actionable in design.

2 Method

To begin positioning “playgrounds for dialogue” as a new genre of museum-related HCI, we are developing it as a *strong concept* [10] – a form of generative, intermediate-level knowledge that resides on an abstraction level above particular instances. We are building this strong concept from the bottom up, by assembling (and later analyzing) a portfolio of design exemplars that we believe reflect what “playgrounds for dialogue” might be. Rather than using our own work as material, we intentionally lean on others’ case studies, allowing us to shape the concept around a broader set of possibilities while foregrounding our ethos of championing the practices of other designers and researchers (in line, e.g., with the proposition of [16]).

Our work began with an unstructured search across online sources – including research articles, social media, blogs, and search engines – to identify museum installations that embodied qualities of social interaction or had the potential to do so and hence aligned with Simon’s idea of museums as “meeting grounds for dialogue.” First, we included: cases from museum-like venues, including museum spaces, galleries, cultural exhibitions, etc.; both technological and non-technological design exemplars, and anything in between, building on a longstanding tradition of leveraging knowledge stemming from analog objects and practices in the design of interactive systems (e.g., see [17] [23] [27]); and existing museum implementations, and conceptual or research-based designs in real spaces. Once we identified exemplars that we thought were fitting, the first author included them in the dataset. The initial dataset included 30 design cases, from which we excluded 6 that were too ambiguously described or focused on virtual experiences beyond museum spaces. That led to a shortlist of 24 design exemplars, whose detailed information was collected and included in a shared digital spreadsheet

(available for download: <https://bit.ly/4rYAkNl>). Our intention was never to create a comprehensive portfolio of all playful museum installations designed to foster social connections among audiences, but rather, from these exemplars, to focus on understanding how specific forms of playful interaction can foster different kinds of social engagement and how these reshape relationships among different visitors in museum spaces.

We then set out to analyze the dataset. We used a combination of close reading [8][25] and reflexive thematic analysis [6] – examining each installation’s affordances, how each one worked, and available evidence about the social interactions it supported, and drawing on our reflexive interpretation to identify themes with potential value for interaction design. Specifically, the first author made the materials tangible by printing a photo of each exemplar that illustrates its functioning and recording the name of the installation, a brief description, and the information sources on the back of the photos. Using the tangible materials and post-it notes, the first and third authors engaged in several reflexive conversations about social mechanisms of the design exemplars to conduct iterative cycles of analysis (Figure 1). When they felt confident in the emerging observations, they shared them with the second author for further contestation and refinement. This analytic process is still ongoing, yet at a stage where a set of early conclusions can already be shared. In the following section, we present these preliminary outcomes, which begin to elucidate the foundations of our “playgrounds for dialogue” strong concept: a set of social affordances that might sustain such “playgrounds for dialogue” and important considerations associated with each social affordance.

3 In-Progress Findings: Social Affordances with Relevant Design Considerations

Here, we report on the early in-progress findings from a reflective analysis of our portfolio of exemplars: First, through looking at our granular annotation of the portfolio, we identify three social affordances that can support playful forms of social connections among diverse visitors around museum content, illustrating how each manifests in practice through selected exemplars from the portfolio; Then, we begin to unearth important design considerations that might assist designers in developing interactive technologies

that present these social affordances. We bring forth these considerations interwoven with the three social affordances described below, showing the interdependencies between social affordances and design considerations.

3.1 Social Affordance #1: People Create Individual Content that Contributes to a Collective Piece.

In examining our collection of exemplars, we realized that a common way for installations to support social connections was allowing visitors to “create individual content that contributes to a collective piece.” That meant empowering visitors to create personalized content, rather than simply consuming content curated by museum institutions, and to append it to a shared canvas where others can see, respond to, or sometimes co-create. Across our portfolio, museum installations support these kinds of creative processes in diverse forms. Some exemplars encourage people to create drawings and contribute them to a shared, larger artwork. Sketch Flight [35] is a good example, in which participants begin by coloring an airplane, a butterfly, or a hawk on provided paper and then display their creation on a shared screen by scanning their drawings and appending them to a larger landscape where the drawings of other people are already displayed. Some exemplars also allow people to leave text-based expressions on a collective display, such as MMAC Wish Tree [15], where people can leave their written wishes and submit them to be displayed on a large, tree-shaped screen. And even some exemplars enable people to contribute representations of their own presence or likeness. For example, Reface [12] enables participants to photograph their faces, which the system then combines with others’ faces and displays on a large-scale display. These exemplars feature stress-free and playful forms, providing visitors with joyful experiences of involvement in a social act of co-construction.

While these exemplars motivate individual content creation, we found that facilitating social exchange depends less on what content people create alone than on in which ways individual content is shared with others and influences others, i.e., *how individual creation is bridged to collective contribution*. Different ways of this bridging result in varying degrees of social exchange. For example, MMAC Wish Tree [15] allows visitors to write wishes on a kiosk and then submit them sequentially to a publicly visible display. In contrast, Color Shake [18] enables participants to co-create graffiti directly on a wall-sized LED display in real time. By supporting simultaneous participation, this experience not only makes individual creations visible but also extends opportunities for co-presence and collaboration. These exemplars motivate consideration of the temporality of participation, both between individual creation and display and between different participants’ involvement – *How to shorten, or even eliminate, time lags between an individual creating and displaying a contribution to a shared piece? How to support synchronous creation among multiple users?*

Looking at our granular annotations associated with exemplars featuring this social affordance, we saw that transparency in authorship should be given important consideration, as participants are positioned as “creators” rather than merely “users.” Beyond [37] is an example: a mixed-reality museum experience that allows

visitors to collect virtual characters from real physical collections and share them as AR filters through photos and videos on social media. Using social media platforms can make authorship explicit while still allowing individuals to control what information they disclose. Which leads us to consider: *how to clarify creators’ identities appropriately? To what extent, and in what forms, can authorship information be displayed without compromising personal privacy?*

Interestingly, some installations extend interaction after extending individual content into a collective piece. Sketch Flight [35] allows users not only to share their creations on a collective screen but also to interact with them further. They can control their creation on a tablet from their perspective and touch the creation on screen to modify its motion. Similarly, Sketch Kusaji Odori [34] enables users to create characters and interact with any character on the shared screen by touch to elicit their surprise actions. These installations prolong engagement and create additional opportunities for shared play. Designers might thus consider subsequent interaction – *how to prolong the interaction after displaying the individual contributions to a collective piece? And how to further foster social connectedness through subsequent interaction with existing collective contributions?*

3.2 Social Affordance #2: People Act Individually, Yet in Co-Presence to Shape a Shared Experience.

Our second social affordance stems from design exemplars in which people act individually, without explicit collaboration, yet through co-presence they collectively shape a shared experience. Beach [28] is a good example: it is an indoor “ocean” landscape composed of white plastic props where visitors can play carelessly, “swimming” in the “ocean,” sitting on a bench on the “beach,” and so on. Each person’s presence and activity in the scene collectively shape the installation into a vivid, beach-like environment, while the evolving environment shapes visitors into co-players.

In such exemplars, experiences can change relationships among people – fragments of each person’s interactions within an experience can be integrated into a collective experience. For example, Light Ball Orchestra [32] is a spatial installation featuring numerous light balls of varying sizes scattered throughout. People can use their bodies to freely touch and roll balls to change their colors and produce sound in this space. More and more, the sounds generated by different people form an orchestral effect. It was designed by TeamLab, a well-known international interdisciplinary art collective. One of their salient design concepts is “[to] make the presence of others a positive experience” [31], aiming to blur the line between audience and artwork and connect participants through shared experiences. The second social affordance we identified aligns with this design concept. At the same time, our research aims to unearth the underlying qualities of this affordance and to help designers intentionally create experiences that present it. From our analysis, we began to surface early-stage design considerations for interactive installations intended to facilitate social engagement through this affordance.

We noticed that individual action and its impact are important design considerations for this affordance. For example, in Sliding through the Fruit Field [33], people can slide down an incline freely

to interact with different balls: light blue balls represent water, yellow balls represent honeybees, and black balls represent contaminated trash that causes seeds to disappear. The collisions of these diverse balls triggered by individuals' actions can affect plant growth. This leads to a prompt consideration: *How to highlight the uniqueness of each individual action while allowing multiple impacts to shape into a collective experience?*

Further, we found that correlation among individual actions, overall experience, and others' actions strongly affects social connectedness. For example, Social Circles [29] is a dance-floor-sized installation where each visitor is encased in a colored bubble on the floor. When people's bubbles bump into each other's, the circles get bigger and change colors. Here diverse individuals' physical movements shape the overall experience, and synchronized changes foster a stronger connection between individuals and others. This exemplar prompt considers *how to form a flexibly coupled system among each individual and the overall experience.*

3.3 Social Affordance #3: People Gather Around to Collaborate and Act Together.

Across our portfolio, we found a recurring way in which visitors intentionally come together to collaborate and co-act around museum content or activities. Our exemplars that present this affordance take diverse forms. In some installations, people can co-perform together, such as Telepresence Charades [22], which is a voiceless, touchless telepresence game deployed across museums. It enables two people at different locations to explain museum collections to one another through silent pantomime or drawing. In other exemplars, people co-play together. Never Let Me Go [19] is an example. It is a two-player experience in a museum setting in which one participant plays the role of a "Controller," who can send voice messages (e.g., instructions, questions, and commands) in real time via a web interface to another participant who plays the role of an "Avatar." Also, people can share a media experience together. For example, Eaves [3] is a handheld device that allows visitors to listen to the same narration together by physically placing one device on top of another.

We found that the play structure [2] in those exemplars ranges from loosely formulated to rigidly defined. Some of these exemplars with game-like structure provide participants with clear roles and goals, making them understand how to collaborate and co-act with others. In Telepresence Charades [22], participants are aware of their roles as "performer" and "answerer," and collaborate

through improvisational play to explain museum collections. Similarly, GIFT [30] explicates the role configuration of participants. It is a smartphone social application that allows two visitors to take on the roles of "gift-maker" and "gift-receiver" to create and exchange digital gifts during their museum visit. In these kinds of experiences, we noticed that participant configuration emerges as a central consideration, including both role configuration and the number of participants. In particular, *how to integrate role configuration into the experience design meaningfully? How can designers determine and support the appropriate number of participants and roles?*

By contrast, some exemplars enable more open-ended play without rules, or with ambiguous ones. For example, in Haunted Play House [36], visitors can literally twist and turn paintings and become part of the artwork by entering the internal space. They can freely shift between the roles of "performer" and "spectator," possessing greater autonomy in how they engage with others and with the experience. In this case, designers might consider *how to provide free-form co-presence for engagement.*

4 Conclusion and Future Work

In this study, we identified three social affordances that can support playful forms of social connections among diverse visitors around museum content. They lay the foundations of our in-progress strong concept: they build on existing design exemplars (and their salient qualities) to begin charting the scope and possible variations of what "playgrounds for dialogue" might be. The first social affordance is allowing people to create personalized content and append it to a collective piece that others can see, respond to, or co-create, enabling them to build social connections without co-presence or collaboration; the second is that people can act individually without explicit collaboration, while through co-presence, they co-shape a shared experience; the third affordance necessitates co-presence and collaboration, where they intentionally gather around to collaborate and co-act around museum content or activities. Together, these social affordances begin to hint at starting points for designing "playgrounds for dialogue": they point toward different ways in which social, communitary, dialogical activity can be supported through playful tech.

To continue making these affordances actionable, we also began to unearth design considerations for each of them. That part of our work is particularly at an early stage: we see the design considerations surfaced here as early, in-progress, and far from

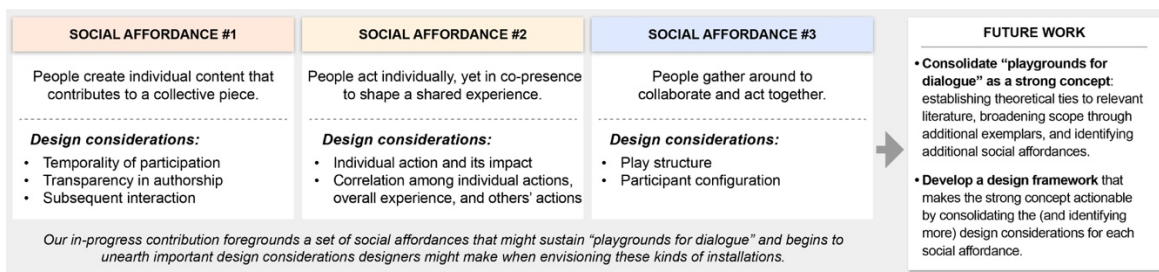


Figure 2: Our in-progress findings from analyzing an annotated portfolio and targeted future work.

consolidated insights that we will deepen as we move forward, potentially leading to the formulation of a design framework.

Figure 2 presents a visual synthesis of our in-progress contribution and our plans for future work. We intend to advance this research from two pathways. On the one hand, we will consolidate “playgrounds for dialogue” as a *strong concept* – establishing theoretical ties to relevant literature of social and play, and broadening scope through additional exemplars to identify more comprehensive social affordances. On the other hand, we expect to develop a design framework that makes the strong concept actionable by consolidating the (and identifying more) design considerations for each social affordance. Overall, our work in that regard is preliminary and will be furthered in future stages of our research, including the very conversations we expect to have with other scholars at CHI.

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